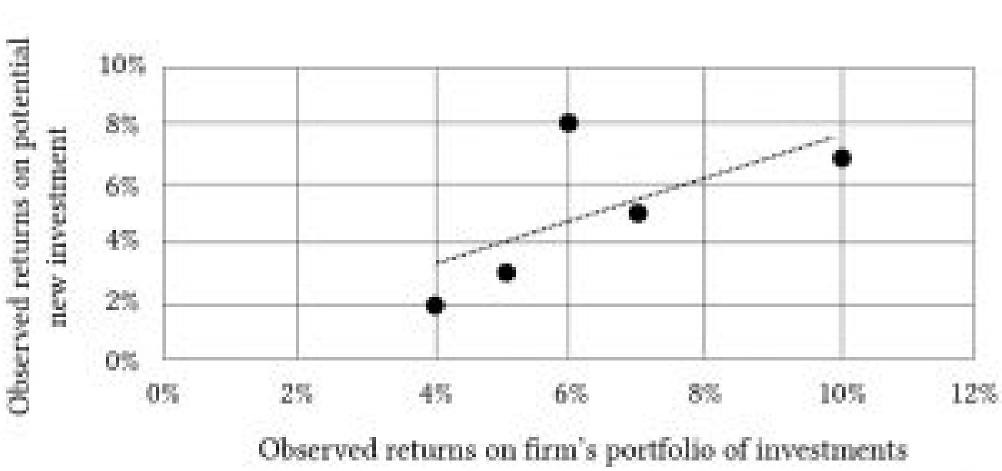
Table 15.6. Observations of Returns on the Firm's Portfolio of Investments r_t^p and on a Potential New Investment (a Challenger).

Time t	Observed returns on the firm's portfolio over time r_t^p	Observed returns on a potential new investment for the firm's r_t^j
2012	10%	7%
2013	6%	8%
2014	7%	5%
2015	3%	2%
2016	5%	3%

Another way to represent the two rates of return measures and their relationship to each other is to represent them in a two dimensional scatter graph.

We may visually observe how the two sets of rates of return move together by drawing a line through the points on the graph in such a way as to minimize the squared distance from the point to the line. Our scatter graph is identified as Figure 15.3.

Figure 15.3. Scatter Graph of Returns on the Firm's Portfolio of Investments and Returns on the Potential New Investment



The relationship between the returns on the new investment and the firm's portfolio can be expressed as:

(15.42) $$r_t^j = a + \beta r_t^j + \epsilon_t$$